



CYS233 Project

THree-Room Office Setup

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Introduction

Background

This project focuses on designing a reliable and efficient network topology using Cisco Packet Tracer 8.2 to establish seamless communication between servers, devices, and interconnected systems. The aim is to create a network infrastructure that supports smooth data exchange, strong connectivity, and improved operational efficiency across various organizational units

Objective of the Network

The primary objective of this project is to design a robust and scalable network that ensures uninterrupted communication and smooth data exchange across various organizational units. The network aims to integrate wired and wireless devices seamlessly, enabling compatibility and efficient operation within the infrastructure. Additionally, it focuses on implementing reliable data routing and secure sharing mechanisms to facilitate effective collaboration between different departments. By utilizing key components such as routers, switches, and other essential devices, the design seeks to optimize network performance while ensuring scalability to accommodate

Topology Requirements

The topology includes the following elements:

Table 1: Topology elements

PC	8
Laptop	5
Printer-PT	3
CISCO Switch	3
CISCO Router	3
IP Phone	4
Server	3
DLS Modem	3
Cloud-PT	3
Access PT	3

Topology

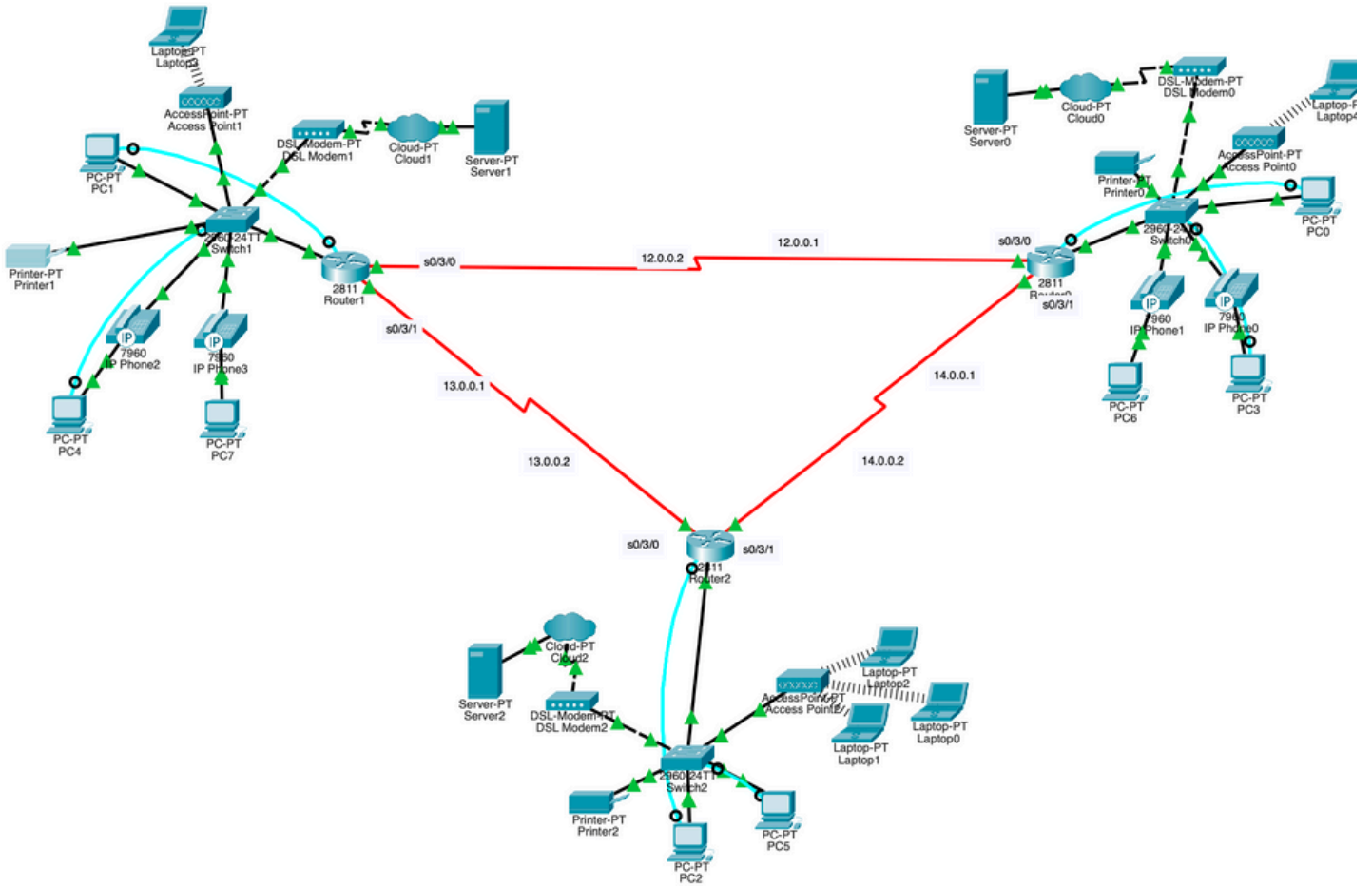


Figure 1: Topology image

Addressing/Ports Table

Device	Interface	IPv4 Address	Subnet Mask	Default GateWay
Router 0 (R0)	FastEthernet0/0	192.168.1.1	255.255.255.0	N/A
	Serial0/3/0	12.0.0.1	255.255.255.252	N/A
	Serial0/3/1	14.0.0.1	255.255.255.252	N/A
Router 1 (R1)	FastEthernet0/0	192.168.2.1	255.255.255.0	N/A
	Serial0/3/0	12.0.0.2	255.255.255.252	N/A
	Serial0/3/1	13.0.0.1	255.255.255.252	N/A
Router 2 (R2)	FastEthernet0/0	192.168.3.1	255.255.255.0	N/A
	Serial0/3/0	13.0.0.2	255.255.255.252	N/A
	Serial0/3/1	14.0.0.2	255.255.255.252	N/A
IP PHONE 0 1112	Vlan1	192.168.1.4	255.255.255.0	192.168.1.1
IP PHONE 1 1111	Vlan1	192.168.1.3	255.255.255.0	192.168.1.1
IP PHONE 2 2222	Vlan1	192.168.2.6	255.255.255.0	192.168.2.1
IP PHONE 3 2223	Vlan1	192.168.2.5	255.255.255.0	192.168.2.1
Printer 0	FastEthernet0	192.168.1.8	255.255.255.0	192.168.1.1
Printer 1	FastEthernet0	192.168.2.9	255.255.255.0	192.168.2.1
Printer 2	FastEthernet0	192.168.3.5	255.255.255.0	192.168.3.1

Addressing/Ports Table

Device	Interface	IPv4 Address	Subnet Mask	Default GateWay
PC1	FastEthernet0	192.168.2.7	255.255.255.0	192.168.2.1
PC2	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.4	255.255.255.0	192.168.2.1
PC4	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC5	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC6	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.1
PC7	FastEthernet0	192.168.3.4	255.255.255.0	192.168.3.1
PC8	FastEthernet0	192.168.3.3	255.255.255.0	192.168.3.1
Server0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
Server1	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
Server2	FastEthernet0	192.168.3.2	255.255.255.0	192.168.3.1
Cloud-PT0	Modem4	N/A	N/A	N/A
	Ethernet6	N/A	N/A	N/A
Cloud-PT1	Modem4	N/A	N/A	N/A
	Ethernet6	N/A	N/A	N/A
Cloud-PT2	Modem4	N/A	N/A	N/A
	Ethernet6	N/A	N/A	N/A

Router's Configuration

Methods

Table 2: Router Configurations Explanation

R(config)# hostname <NAME>	Set the host-name for routers.
R(config)# enable secret <SECRET>	Set a password to protect privileged EXEC mode.
R(config)# line console 0 R(config)# password <SECRET> R(config)# login	Set a password to protect the console interface.
R(config)# line vty 0 4 R(config-line)# password <SECRET> R(config-line)# login	Set a password to protect the VTY (Telnet/SSH) interface.
R(config)# service password-encryption	Enable password encryption service.
R(config)# banner motd # <MESSAGE> #	Setting an motd banner.
R(config)# interface <INTERFACE> R(config-if)# ip address <IP> <MASK> R(config-if)# no shutdown	Assign an IP address to an interface and enable it.
R(config)# ip route <NETWORK> <MASK> <NEXT-HOP>	Add a static route to the routing table.
R(config)# copy running-config startup-config	Save the current configuration to the startup configuration file

Applied Configuration

1. Enable secret: <CLASS>
2. Line console and vty password: <AAAA>

(Router0):

Network Address

192.168.2.0/24 via 12.0.0.2 192.168.3.0/24 via 14.0.0.2

192.168.2.0/24 via Serial0/3/0 192.168.1.0/24 via Serial0/3/1

Figure 2 : R0 Network Address

Device Name: Router0					
Custom Device Model: 2811 IOS15					
Hostname: R0					
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/0	Up	--	192.168.1.1/24	<not set>	0002.4A93.3301
FastEthernet0/1	Down	--	<not set>	<not set>	0002.4A93.3302
Serial0/3/0	Up	--	12.0.0.1/30	<not set>	<not set>
Serial0/3/1	Up	--	14.0.0.1/30	<not set>	<not set>
Vlan1	Down	1	<not set>	<not set>	0090.2B8E.E99B
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router0					

Figure 3 : R0 Interface

(Router1):

Network Address

192.168.1.0/24 via 12.0.0.1 192.168.3.0/24 via 13.0.0.2

192.168.1.0/24 via Serial0/3/0 192.168.3.0/24 via Serial0/3/0

Figure 4 : R1 Network Address

Device Name: Router1					
Custom Device Model: 2811 IOS15					
Hostname: R1					
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/0	Up	--	192.168.2.1/24	<not set>	0001.4321.7601
FastEthernet0/1	Down	--	<not set>	<not set>	0001.4321.7602
Serial0/3/0	Up	--	12.0.0.2/30	<not set>	<not set>
Serial0/3/1	Up	--	13.0.0.1/30	<not set>	<not set>
Vlan1	Down	1	<not set>	<not set>	00E0.A39C.BEB2
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router1					

Figure 5 : R1 Interface

(Router2):

Network Address

192.168.1.0/24 via 14.0.0.1 192.168.2.0/24 via 13.0.0.1

192.168.1.0/24 via Serial0/3/1 192.168.2.0/24 via Serial0/3/1

Figure 6 : R2 Network Address

Device Name: Router2					
Custom Device Model: 2811 IOS15					
Hostname: R2					
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/0	Up	--	192.168.3.1/24	<not set>	000D.BD99.1201
FastEthernet0/1	Down	--	<not set>	<not set>	000D.BD99.1202
Serial0/3/0	Up	--	13.0.0.2/30	<not set>	<not set>
Serial0/3/1	Up	--	14.0.0.2/30	<not set>	<not set>
Vlan1	Down	1	<not set>	<not set>	000C.85D8.5BB2
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router2					

Figure 7 : R2 Interface

Switch Configuration

Methods

Table 3: Switch Configurations Explanations

Switch(config)# hostname <NAME>	Set the hostname for the switch to <NAME>.
Switch(config)# enable secret <SECRET>	Set the privileged mode password to <SECRET>.
Switch(config)# line console 0 Switch(config-line)# password <SECRET> Switch(config-line)# login	Set the console password to <SECRET> and enable login for console access.
Switch(config)# line vty 0 4 Switch(config-line)# password <SECRET> Switch(config-line)# login	Set the VTY password to <SECRET> and enable login for remote access via Telnet/SSH.
Switch(config)# service password-encryption	Enable password encryption to protect plain-text passwords.
Switch(config)# banner motd #<TEXT>#	Setting an motd banner.
Switch(config)# vlan <VLAN ID> Switch(config-vlan)# name <NAME>	Create VLAN <VLAN ID> and assign the name <NAME>.
Switch(config)# interface range <INTERFACE RANGE> Switch(config-if-range)# switchport mode access Switch(config-if-range)# switchport voice vlan <VLAN ID>	Configure interfaces <INTERFACE RANGE> as access mode and assign VLAN <VLAN ID> for voice traffic.
Switch# copy running-config startup-config	Save the current configuration to the startup configuration file.

Applied Configuration

1. Enable secret: **CLASS**
2. Line console and vty password: **AAAA**

VoIP Configuration

IP Phone 0 :

Figure 8 : IP Phone0 interfaces

```
Device Name: IP Phone0
Device Model: 7960

Port    Link    IP Address    MAC Address
Vlan1   Up      192.168.1.5/24    0009.7CE0.1DB0
Switch  Up      <not set>        0001.C714.9801
PC       Up      <not set>        0001.C714.9802

Gateway: 192.168.1.1
Line Number: 1112

Physical Location: Intercity > Home City > Corporate Office > IP Phone0
```

IP Phone 1 :

Figure 9 : IP Phone1 interfaces

```
Device Name: IP Phone1
Device Model: 7960

Port    Link    IP Address    MAC Address
Vlan1   Up      192.168.1.4/24    0003.E430.D247
Switch  Up      <not set>        00E0.A3A4.C601
PC       Up      <not set>        00E0.A3A4.C602

Gateway: 192.168.1.1
Line Number: 1111

Physical Location: Intercity > Home City > Corporate Office > IP Phone1
```

IP Phone 2 :

Figure 10 : IP Phone2 interfaces

```
Device Name: IP Phone2
Device Model: 7960

Port    Link    IP Address    MAC Address
Vlan1   Up      192.168.2.6/24    0010.1170.86B1
Switch  Up      <not set>        000C.852A.5401
PC       Up      <not set>        000C.852A.5402

Gateway: 192.168.2.1
Line Number: 2222

Physical Location: Intercity > Home City > Corporate Office > IP Phone2
```

IP Phone 3 :

Figure 11 : IP Phone3 interfaces

```
Device Name: IP Phone3
Device Model: 7960

Port    Link    IP Address    MAC Address
Vlan1   Up      192.168.2.5/24    000B.BE06.8667
Switch  Up      <not set>        0001.9676.8E01
PC       Up      <not set>        0001.9676.8E02

Gateway: 192.168.2.1
Line Number: 2223

Physical Location: Intercity > Home City > Corporate Office > IP Phone3
```

PC Configuration

PC1 :

Figure 12 : PC1 GateWayConfiguartion

Gateway/DNS IPv4	
☒ DHCP	
☐ Static	
Default Gateway	192.168.2.1
DNS Server	

Figure 13 : PC1 IP Configuartion

IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	192.168.2.7
Subnet Mask	255.255.255.0

PC2 :

Figure 14 : PC2 GateWayConfiguartion

Gateway/DNS IPv4	
☒ DHCP	
☐ Static	
Default Gateway	192.168.2.1
DNS Server	

Figure 15 : PC2 IP Configuartion

IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	192.168.2.3
Subnet Mask	255.255.255.0

PC3 :

Figure 16 : PC3 GateWayConfiguartion

Gateway/DNS IPv4	
☒ DHCP	
☐ Static	
Default Gateway	192.168.2.1
DNS Server	

Figure 17 : PC3 IP Configuartion

IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	192.168.2.4
Subnet Mask	255.255.255.0

PC Configuration

PC4 :

Figure 18 : PC4 GateWayConfiguartion

Gateway/DNS IPv4	
☒ DHCP	
☐ Static	
Default Gateway	192.168.1.1
DNS Server	

Figure 19 : PC4 IP Configuartion

IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	192.168.1.2
Subnet Mask	255.255.255.0

PC5 :

Figure 20 : PC5 GateWayConfiguartion

Gateway/DNS IPv4	
☒ DHCP	
☐ Static	
Default Gateway	192.168.1.1
DNS Server	

Figure 21 : PC5 IP Configuartion

IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	192.168.1.3
Subnet Mask	255.255.255.0

PC6 :

Figure 22 : PC6 GateWayConfiguartion

Gateway/DNS IPv4	
☒ DHCP	
☐ Static	
Default Gateway	192.168.1.1
DNS Server	

Figure 23 : PC6 IP Configuartion

IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	192.168.1.10
Subnet Mask	255.255.255.0

PC Configuration

PC7 :

Figure 24 : PC7 GateWayConfiguartion

Gateway/DNS IPv4	
☒ DHCP	
☐ Static	
Default Gateway	192.168.3.1
DNS Server	0.0.0.0

Figure 25 : PC7 IP Configuartion

IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	192.168.3.4
Subnet Mask	255.255.255.0

PC8 :

Figure 26 : PC8 GateWayConfiguartion

Gateway/DNS IPv4	
☒ DHCP	
☐ Static	
Default Gateway	192.168.3.1
DNS Server	0.0.0.0

Figure 27 : PC8 IP Configuartion

IP Configuration	
☒ DHCP	
☐ Static	
IPv4 Address	192.168.3.3
Subnet Mask	255.255.255.0

Printers Configuration

Printer0 :

Figure 28 : Printer0 GateWayConfiguartion

Gateway/DNS IPv4	
☐ DHCP	
☒ Static	
Default Gateway	192.168.1.1
DNS Server	

Figure 29 : Printer0 IP Configuartion

IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.1.8
Subnet Mask	255.255.255.0

Printer1 :

Figure 30 : Printer1 GateWayConfiguartion

Gateway/DNS IPv4	
☐ DHCP	
☒ Static	
Default Gateway	192.168.2.1
DNS Server	

Figure 31 : Printer1 IP Configuartion

IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.2.9
Subnet Mask	255.255.255.0

Printer2 :

Figure 32 : Printer2 GateWayConfiguartion

Gateway/DNS IPv4	
☐ DHCP	
☒ Static	
Default Gateway	192.168.3.1
DNS Server	

Figure 33 : Printer2 IP Configuartion

IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.3.5
Subnet Mask	255.255.255.0

Servers Configuration

Server0 :

Figure 34 : Server0 GateWayConfiguartion

Gateway/DNS IPv4	
☐ DHCP	
☒ Static	
Default Gateway	192.168.1.1
DNS Server	

Figure 35 : Server0 IP Configuration

IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.1.2
Subnet Mask	255.255.255.0

Server1 :

Figure 36 : Server1 GateWayConfiguartion

Gateway/DNS IPv4	
☐ DHCP	
☒ Static	
Default Gateway	192.168.2.1
DNS Server	

Figure 37 : Server1 IP Configuration

IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.2.2
Subnet Mask	255.255.255.0

Server2 :

Figure 38 : Server2 GateWayConfiguartion

Gateway/DNS IPv4	
☐ DHCP	
☒ Static	
Default Gateway	192.168.3.1
DNS Server	

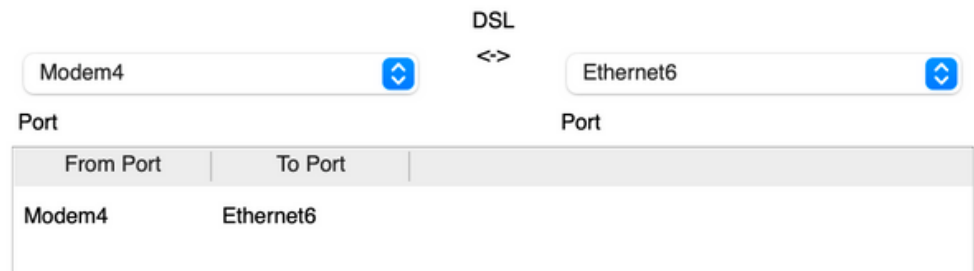
Figure 39 : Server2 IP Configuration

IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.3.2
Subnet Mask	255.255.255.0

DSL Modem Configuration

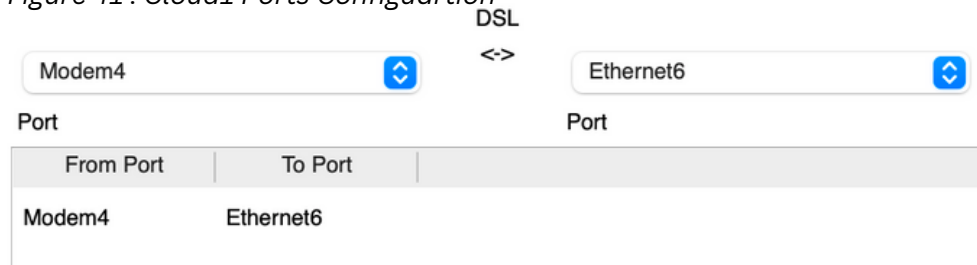
Cloud0 :

Figure 40 : Cloud0 Ports Configuartion



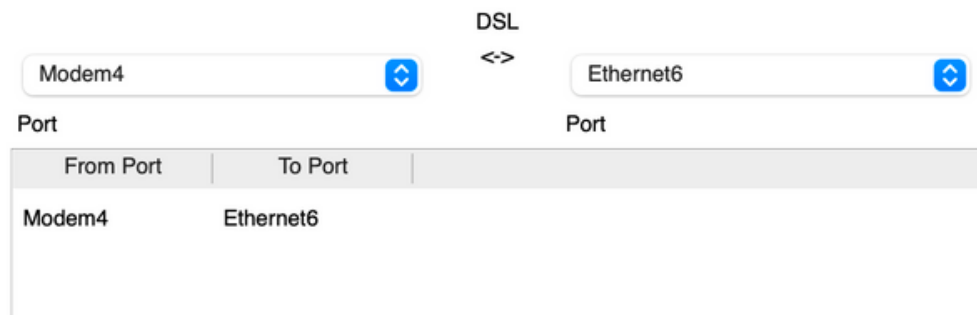
Cloud1 :

Figure 41 : Cloud1 Ports Configuartion



Cloud2 :

Figure 42 : Cloud2 Ports Configuartion



Verification

IP Phones connected:

Figure 43 : VoIP test



Every network is connected (ping is working)

Figure 44 : ping from PC0 to PC1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Physical Address...: 000D.BD30.25ED
    Link-local IPv6 Address...: FE80::200B:DFF:FE30:25ED
    IPv6 Address...: ::
    IPv4 Address...: 192.168.1.2
    Subnet Mask...: 255.255.255.0
    Default Gateway...: ::
    DHCP Servers...: 192.168.1.1
    DHCPv6 IAID...: 192.168.1.1
    DHCPv6 Client DUID...: 00-01-00-01-99-6E-6E-0C-00-0D-BD-30-25-ED
    DNS Servers...: ::
    0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address...: 0001.63CC.002D
    Link-local IPv6 Address...: ::
    IPv6 Address...: ::
    IPv4 Address...: 0.0.0.0
    Subnet Mask...: 0.0.0.0
    Default Gateway...: ::
    0.0.0.0
    DHCP Servers...: 0.0.0.0
    DHCPv6 IAID...: 
    DHCPv6 Client DUID...: 00-01-00-01-99-6E-6E-0C-00-0D-BD-30-25-ED
    DNS Servers...: ::
    0.0.0.0

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=24ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 24ms, Average = 8ms
```

Figure 46 : ping from PC1 to PC0

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Physical Address...: 000C.CFAB.EE48
    Link-local IPv6 Address...: FE80::20C1:CFAB:EE48
    IPv6 Address...: ::
    IPv4 Address...: 192.168.2.3
    Subnet Mask...: 255.255.255.0
    Default Gateway...: ::
    DHCP Servers...: 192.168.2.1
    DHCPv6 IAID...: 192.168.2.1
    DHCPv6 Client DUID...: 00-01-00-01-C6-DD-B2-18-00-0C-CF-AB-EE-48
    DNS Servers...: ::
    0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address...: 0060.3E43.8892
    Link-local IPv6 Address...: ::
    IPv6 Address...: ::
    IPv4 Address...: 0.0.0.0
    Subnet Mask...: 0.0.0.0
    Default Gateway...: ::
    0.0.0.0
    DHCP Servers...: 0.0.0.0
    DHCPv6 IAID...: 
    DHCPv6 Client DUID...: 00-01-00-01-C6-DD-B2-18-00-0C-CF-AB-EE-48
    DNS Servers...: ::
    0.0.0.0

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=20ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=2ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 20ms, Average = 10ms
```

Figure 49 : ping from PC2 to PC1

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=36ms TTL=125
Reply from 192.168.2.3: bytes=32 time=21ms TTL=126
Reply from 192.168.2.3: bytes=32 time=32ms TTL=125
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 36ms, Average = 22ms
```

Figure 45 : ping from PC0 to PC2

```
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.3: bytes=32 time=24ms TTL=126
Reply from 192.168.3.3: bytes=32 time=22ms TTL=126
Reply from 192.168.3.3: bytes=32 time=23ms TTL=126

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 24ms, Average = 23ms
```

Figure 47 : ping from PC1 to PC2

```
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time=31ms TTL=126
Reply from 192.168.3.3: bytes=32 time=34ms TTL=125
Reply from 192.168.3.3: bytes=32 time=3ms TTL=126
Reply from 192.168.3.3: bytes=32 time=33ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 34ms, Average = 25ms
```

Figure 48 : ping from PC2 to PC0

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Physical Address...: 0040.70E5.4A6E
    Link-local IPv6 Address...: FE80::2607:0FF:FE55:4A6E
    IPv6 Address...: ::
    IPv4 Address...: 192.168.3.3
    Subnet Mask...: 255.255.255.0
    Default Gateway...: ::
    DHCP Servers...: 192.168.3.1
    DHCPv6 IAID...: 192.168.3.2
    DHCPv6 Client DUID...: 00-01-00-01-7C-18-45-68-00-60-70-E5-4A-6E
    DNS Servers...: ::
    0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address...: 00E0.A3BA.054C
    Link-local IPv6 Address...: ::
    IPv6 Address...: ::
    IPv4 Address...: 0.0.0.0
    Subnet Mask...: 0.0.0.0
    Default Gateway...: ::
    0.0.0.0
    DHCP Servers...: 0.0.0.0
    DHCPv6 IAID...: 
    DHCPv6 Client DUID...: 00-01-00-01-7C-18-45-68-00-60-70-E5-4A-6E
    DNS Servers...: ::
    0.0.0.0

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=31ms TTL=126
Reply from 192.168.1.2: bytes=32 time=23ms TTL=126
Reply from 192.168.1.2: bytes=32 time=25ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 31ms, Average = 20ms
```



Conclusion

The project emphasizes the vital role of well-structured networks in modern organizations for ensuring seamless communication and efficient data management. A properly designed network enables smooth collaboration, enhances productivity, and lays a solid foundation for operational success. This network will not only meet the current demands of the organization but also adapt to future technological advancements and growth opportunities.